

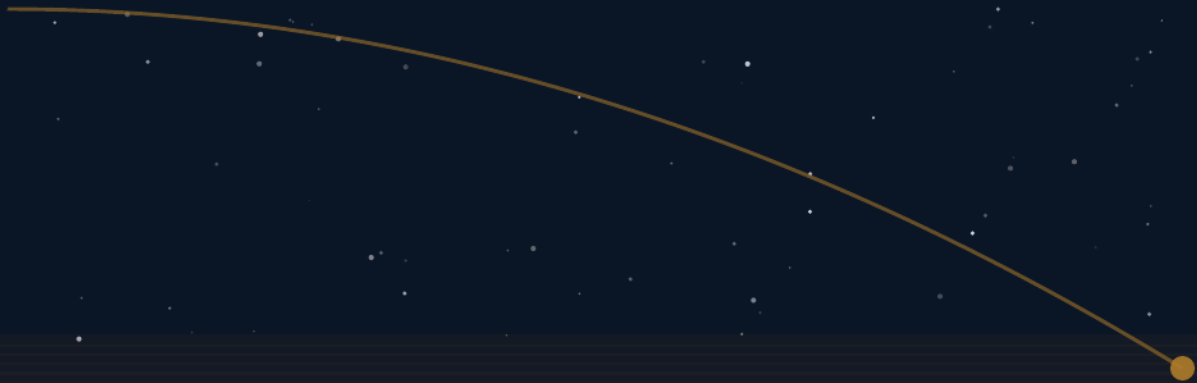
MAGELLICA PRESS · ASTRONOMY

Twilight Duration

Why It Stays Light After the Sun Has Gone Down

STUDENT STUDY GUIDE

Interactive slide-quiz · Quiz ID TWILIGHT01



BEFORE YOU BEGIN

Quick Reference

What this exercise is about

Twilight lasts as long as the Sun takes to sink 18° below the horizon. How long that takes depends on how far the Sun's path leans away from the vertical. Upright path, brisk twilight; leaning path, long twilight. Everything in the three pages follows from that one sentence.

Learning objectives

- Explain why twilight lasts longer at high latitude even though the Sun always tries to sink the same 18° .
- Use the equinox rule that the Sun's path leans away from the vertical by the observer's latitude.
- Distinguish civil, nautical and astronomical twilight, and say what each one is good for.
- Account for a blank timer: a stage that never begins/ends because the Sun never gets deep enough.
- Explain the midnight Sun at Fort Yukon, and why the western view has to be abandoned to see it.

Key terms

Twilight	The interval between sunset and the moment the Sun reaches 18° below the horizon. The vertical drop is the same everywhere; the time it takes is not.
Lean from the vertical	How far the Sun's descending path tips away from straight down. On an equinox it equals the observer's latitude. Small lean, brisk twilight; large lean, long twilight.
Civil twilight	Sunset to 6° below. You can still read outside and the horizon is clear.
Nautical twilight	6° to 12° below. Horizon and bright stars are visible at the same moment, which is what a sextant sighting needs.
Astronomical twilight	12° to 18° below. It looks like night, but the sky is still lit enough to spoil a faint-galaxy exposure.
Midnight Sun	A day on which the Sun never sets. With no sunset there is no twilight to time, which is why every box on the timer panel stays blank.
Azimuth	The compass direction of the Sun on the horizon. On an equinox it is due west, 270.0° , from every latitude.

Numbers worth remembering

18°	Depth below the horizon at which twilight ends and true night begins
6° bands	Width of each of the three twilight stages
72 min	Total twilight at the equator on an equinox — the shortest case
98 min	Total twilight at Chicago, 41.9°N , on an equinox
119 min	Total twilight at London, 51.5°N , on an equinox
41.9°	Chicago's lean from the vertical on an equinox — the same as its latitude
about 15°	Deepest the Sun gets below London's horizon at midsummer — short of 18°

48.6° / 54.6° / 60.6°

Summer-solstice latitudes above which astronomical, then nautical, then civil twilight stop happening

66.6°

Latitude above which the Sun stops setting at midsummer — the Arctic Circle

PART 1 REVIEW

The Angle of the Sun's Path

Animation setup: Vernal Equinox · viewpoint W · latitude slider to 0, then 35, then Chicago, then London

Twilight is the interval between sunset and the moment the Sun reaches 18° below the horizon. That vertical drop requirement never changes. What changes from place to place is how long the Sun takes to cover it, if it can.

At the equator, on an equinox day, the Sun goes down like a stone, leaning 0° from the vertical, and total twilight runs 72 minutes. Move north and the path tips over: the Sun slides down and to the right as it sets, covering the same 18° more slowly. At 35°N it takes 89 minutes, at Chicago 98, at London 119.

The rule behind it is tidy. On an equinox day, and only on an equinox day, the lean away from the vertical equals your latitude: 0° at the equator, about 42° at Chicago, about 51.5° at London. The lean grows with latitude and twilight stretches to match.

One detail matters for Part 3. On an equinox day, the Sun sets due west for every latitude — 270.0° on the azimuth diagram, 0° off due west. Only the steepness of the descent changes, never the direction.

PART 2 REVIEW

The Three Stages of Twilight

Animation setup: Vernal Equinox · location Chicago · viewpoint W · watch the timer card

Twilight is not one seamless event. Astronomers split it into three stages, each a 6° band of the Sun's descent, and the timer panel tracks all three separately.

Civil twilight runs from sunset to 6° down. It is the useful part of the evening: you can still read outside, color is still discernable, the horizon is clear. Nautical twilight carries on to 12° down, and the name is literal — a navigator can still make out the horizon while the brightest stars are out, and a sextant sighting needs both during that time. Astronomical twilight runs to 18° down. It looks like night during that interval, but the sky is still faintly lit, enough to interfere with a telescope hunting faint galaxies.

At the equator the three stages come out equal, 24 minutes each, because a Sun dropping straight down crosses equal vertical bands at an equal rate. Chicago is nearly equal, 32, 33 and 33. Push north to 70° and these intervals diverge badly: 71, 79, then 109. The later bands stretch the most, because the further the Sun sinks at high latitude, the shallower its path becomes.

PART 3 REVIEW

When Twilight Refuses to End

Animation setup: Summer Solstice · London, then Chicago, then Fort Yukon · viewpoint W, then NW, then SW for the winter solstice at the end

Lean the path far enough from vertical and twilight stops ending. On the summer solstice, Chicago still has all three stages, 40, 44 and 54 minutes, two hours and eighteen in total.

London does not. Its astronomical timer stays blank, because the Sun never gets more than about 15° below the horizon and needs 18° to bring on astronomical darkness. There is no fully dark sky in London for weeks around midsummer; the sky at one in the morning is deep blue, not black.

Fort Yukon, at 66.6°N , has every box blank, for a different reason: there is no sunset at all. The Sun circles the observer without touching the horizon — hence the midnight Sun. Switch the viewpoint from West to Northwest and the reason becomes clear: the Sun swings around to the north instead of dropping. Looking west, you were watching the wrong part of the sky.

Each twilight stage gives out at its own latitude. On the summer solstice astronomical twilight disappears above 48.6° , nautical above 54.6° , civil above 60.6° , and the Sun stops setting above 66.6° , the Arctic Circle.

Now set the Winter Solstice at Fort Yukon with the viewpoint Southwest. Now the Sun never rises, and yet the sky visibly brightens around midday and the timers show real figures. Polar night is not a solid block of darkness. It is a long twilight that almost, but never quite, becomes day.

TWELVE QUESTIONS, FOUR PER PAGE

Study Questions

Part 1 — The Angle of the Sun's Path

1. Twilight always ends when the Sun reaches the same depth below the horizon. What is that depth, and why does twilight still last different amounts of time at different latitudes?
2. On an equinox, how do you determine the lean of the Sun's path from the vertical for any observer? Give the value for the equator, for Chicago and for London.
3. A student runs the animation on the equinox at the equator, then at London, and reports 72 minutes and 119 minutes of total twilight. Explain the difference in terms of the Sun's path.
4. On an equinox, in what direction does the Sun set as seen from the equator, from Chicago and from London? What does change between those three places?

Part 2 — The Three Stages of Twilight

1. Name the three stages of twilight in order and give the range of solar depth each one covers.
2. Why is nautical twilight the stage a navigator wants when taking a sextant sighting?

3. At the equator on an equinox the three stages last about 24 minutes each. Explain why they come out equal, and why the same is not true at 70°N.
4. What is happening in the sky during astronomical twilight, and why does an observatory care about the difference between 12° and 18°?

Part 3 — When Twilight Refuses to End

1. At London on the summer solstice the astronomical twilight timer stays blank. What does a blank timer mean here, and what is the physical reason for it?
2. At Fort Yukon on the summer solstice every timer is blank, and the reason is different from London's. What is it, and what does switching the viewpoint to NW reveal?
3. Set the Winter Solstice at Fort Yukon. The Sun never rises, yet the sky brightens around midday and some timers show figures. Explain what is happening.

DISCUSSION, NO CALCULATION

Challenge Questions

- C1** Twilight at the equator is the same 72 minutes on every day of the year, while Chicago's swings from 98 minutes at the equinoxes to 138 at midsummer. What does that tell you about how the Sun's path changes through the year at each place?
- C2** A pilot, a yacht navigator and an observatory director each ask when it gets dark. Which stage of twilight matters to each, and why would a single answer fail all three?
- C3** Fort Yukon sits essentially on the Arctic Circle. Predict what happens to the midnight Sun and to polar night a few hundred miles further north, and say what you would expect the timer panel to show there.
- C4** Some places describe midsummer nights with a name of their own — the white nights of St Petersburg, for example. Using what the animation shows, explain what physically distinguishes such a night from an ordinary dark one, and roughly what latitude a city needs for it.